Quantum Computing
Professor Gopal Pandurangan

Scribe Author: Armaan Amatya

Contents

Lecture of September 1, 2026 **2**

1 Query Complexity for Search **2**
1.1 Monte Carlo and Las Vegas Algorithms 2
1.2 Random Sampling 2

2 Coupon Collector Problem **3**

3 High Probability Bound **4**

4 Lower Bound for Randomized Algorithms **5**

5 Chernoff Bounds **5**

Lecture of September 3, 2026 **7**

6 Markov's Inequality **7**

7 Chebyshev's Inequality **8**

8 Chernoff Bound **8**

9 Search **9**

10 Birthday Paradox **9**
10.1 Expected Number of Collisions 10

11 Quantum Birthday Paradox **11**

Lecture Date: September 1, 2026

Lecture Topic: Query Complexity of Search, Coupon Collector Problem, Lower Bound for Randomized Algorithms, Chernoff Bounds

In this lecture, we study the query complexity of search for classical randomized algorithms. We analyze random sampling, relate it to the coupon collector problem, state a lower bound for randomized search algorithms using Yao's minmax principle, and introduce Chernoff bounds.

1 Query Complexity for Search

We are given an array of n slots (Figure 1). The array can only be accessed through *queries*: a query to a slot x returns the value $f(x)$. In the *search* problem, the goal is to find the slot x such that

$$f(x) = 1.$$

The cost of an algorithm is its *query complexity*, i.e., the number of queries that it makes.

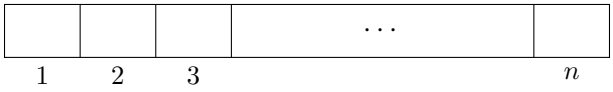


Figure 1: An array of n slots. We want to find the slot x with $f(x) = 1$.

1.1 Monte Carlo and Las Vegas Algorithms

Definition 1 ▶ Monte Carlo and Las Vegas Algorithms

A **Monte Carlo** algorithm is a randomized algorithm whose error probability is > 0.
A **Las Vegas** algorithm is a randomized algorithm whose error probability is $= 0$.

1.2 Random Sampling

Consider random sampling *with replacement*: in every step we query a slot that is chosen uniformly at random among the n slots. For every query,

$$\mathbb{P}(\text{Success}) = \frac{1}{n}.$$

The number of queries until the first success is a *geometric* random variable with parameter $1/n$. Hence

$$\mathbb{E}[\text{queries}] = n.$$

If random sampling is stopped after a fixed number of queries, then it is a Monte Carlo algorithm, since it can fail to find the slot. If we keep sampling until the slot is found, then it is a Las Vegas algorithm, and

$$\mathbb{E}[\text{queries until find}] = n.$$

2 Coupon Collector Problem

Suppose that we want to distinguish 2 binary arrays (Figure 2). To do this with random sampling, all the n slots have to be accessed. The number of queries needed by random sampling (with replacement) to access all the n slots is the *coupon collector problem*. We show that

$$\mathbb{E}[\text{queries}] \leqslant \mathcal{O}(n \log n) \quad \text{and} \quad \mathbb{E}[\text{queries}] \geqslant \Omega(n \log n),$$

i.e., $\mathbb{E}[\text{queries}] = \Theta(n \log n)$.

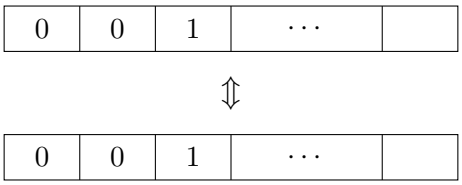


Figure 2: To distinguish 2 binary arrays.

Theorem 1 ▶ Coupon Collector

Let X be the number of queries made by random sampling (with replacement) until all the n slots are accessed. Then

$$\mathbb{E}[X] = nH_n = \Theta(n \log n),$$

where $H_n = \sum_{i=1}^{n} \frac{1}{i}$ is the nth harmonic number.

Proof. Write

$$X = X_1 + X_2 + \cdots + X_n,$$

where X_i is the number of queries made to access a new slot, after $i-1$ different slots have already been accessed. When $i-1$ slots have been accessed, a query accesses a new slot with probability $\frac{n-(i-1)}{n}$. Hence X_i is a geometric random variable,

$$X_i \sim G\left(\frac{n-(i-1)}{n}\right), \qquad \mathbb{E}[X_i] = \frac{n}{n-(i-1)}.$$

By linearity of expectation,

$$\begin{aligned}
\mathbb{E}[X] &= \mathbb{E}[X_1 + X_2 + \cdots + X_n] \\
&= \sum_{i=1}^{n} \mathbb{E}[X_i] \\
&= \sum_{i=1}^{n} \frac{n}{n-(i-1)} \\
&= n \sum_{i=1}^{n} \frac{1}{n-i+1} \\
&= nH_n.
\end{aligned}$$

Since $H_n = \Theta(\log n)$, we get $\mathbb{E}[X] = \Theta(n \log n)$. □

3 High Probability Bound

The following was given in class as an exercise.

Exercise 1. Show that

$$\mathbb{P}(\text{any slot is accessed} \leqslant c \log n \text{ times}) \leqslant \frac{1}{n^2}.$$

Solution.

The expectation of the number of queries is $\Theta(n \log n)$. We now show that $3n \log n$ queries are enough with high probability. Here log is the natural logarithm.

Theorem 2

If random sampling makes $3n \log n$ queries, then

$$\mathbb{P}(\text{fail}) \leqslant \frac{1}{n^2}, \qquad \text{i.e.,} \qquad \mathbb{P}(\text{Success}) \geqslant 1 - \frac{1}{n^2},$$

where "fail" means that some slot is not accessed.

Proof. Let B_i be the event that slot i is not accessed. A query does not access slot i with probability $1 - \frac{1}{n}$. Using $1 - x \leqslant e^{-x}$,

$$\begin{aligned}
\mathbb{P}(B_i) = \mathbb{P}(\text{slot } i \text{ not accessed}) &\leqslant \left(1 - \frac{1}{n}\right)^{3n \log n} \\
&\leqslant e^{-\frac{1}{n} \cdot 3n \log n} \\
&= e^{-3 \ln n} \\
&= \frac{1}{n^3}.
\end{aligned}$$

By the union bound,

$$\mathbb{P}\left(\bigcup_i B_i\right) \leqslant \sum_{i=1}^{n} \mathbb{P}(B_i) = \frac{n}{n^3} = \frac{1}{n^2}.$$

□

4 Lower Bound for Randomized Algorithms

Theorem 3

Any Las Vegas randomized algorithm for search needs $\Omega(n)$ expected queries.

The lower bound is shown using Yao's minmax principle.

Yao's Minmax Principle

Let A^{rand} be a randomized algorithm, and let $\mathbb{E}\left[A^{\text{rand}}\right]$ be its expected number of queries. Let μ be a probability distribution over the inputs, and let D_μ^{best} be the best deterministic algorithm for inputs that are chosen according to μ. Then

$$\mathbb{E}\left[A^{\text{rand}}\right] \geqslant \max_\mu \mathbb{E}_\mu\left[D_\mu^{\text{best}}\right] \geqslant \Omega(n).$$

The quantity $\mathbb{E}_\mu\left[D_\mu^{\text{best}}\right]$ is called the *distributional complexity*.

5 Chernoff Bounds

Let's say that a coin gives heads (H) with probability $\frac{1}{2}+\varepsilon$ and tails (T) with probability $\frac{1}{2}-\varepsilon$. If we toss the coin n times, then

$$\mathbb{E}[\text{Heads}] = n\left(\frac{1}{2}+\varepsilon\right) = \frac{n}{2}+n\varepsilon.$$

Chernoff bounds show that the number of heads is close to its expectation, so that

$$\mathbb{P}\left(\text{Heads} < \frac{n}{2}\right) \sim \text{very small}.$$

In general, let

$$X = \sum_{i=1}^n X_i, \qquad X_i = \begin{cases} 1 & \text{with probability } p_i, \\ 0 & \text{with probability } 1-p_i, \end{cases}$$

where the X_i are independent Bernoulli random variables. If $p_i = p$ for all i, then

$$\mathbb{E}[X] = np, \qquad \mathbb{V}[X] = np(1-p).$$

The distribution of X is concentrated around its mean np (Figure 3).

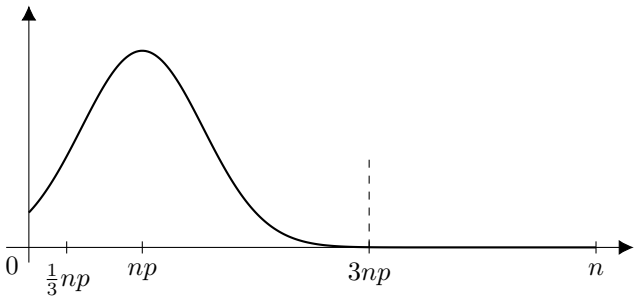


Figure 3: The distribution of X is concentrated around the mean np.

Theorem 4 ▶ Chernoff Bounds

Let $X = \sum_{i=1}^{n} X_i$ be as above and let $\mu = \mathbb{E}[X]$.

- Upper tail (UI): for $0 < \delta < 1$,

$$\mathbb{P}(X \geqslant (1+\delta)\mu) \leqslant e^{-\frac{\mu\delta^2}{3}}.$$

- For $R \geqslant 6\mu$,

$$\mathbb{P}(X \geqslant R) \leqslant \frac{1}{2^R}.$$

- Lower tail (LI): for $0 < \delta < 1$,

$$\mathbb{P}(X \leqslant (1-\delta)\mu) \leqslant e^{-\frac{\mu\delta^2}{2}}.$$

Lecture Date: September 3, 2026

Lecture Topic: Markov, Chebyshev and Chernoff Inequalities, Search, Birthday Paradox

In this lecture, we state and prove Markov's inequality and Chebyshev's inequality, and we state the Chernoff bound. We then look at the query complexity of search for classical and quantum algorithms, and we analyze the birthday paradox.

6 Markov's Inequality

Markov's inequality uses only the expectation $\mathbb{E}[X]$ of the random variable, which is also called the *first moment*.

Theorem 5 ► Markov's Inequality

Let $X \geqslant 0$ be a random variable. Then, for any $a > 0$,

$$\mathbb{P}(X \geqslant a\,\mathbb{E}[X]) \leqslant \frac{1}{a}.$$

Equivalently, for any $b > 0$,

$$\mathbb{P}(X \geqslant b) \leqslant \frac{\mathbb{E}[X]}{b}.$$

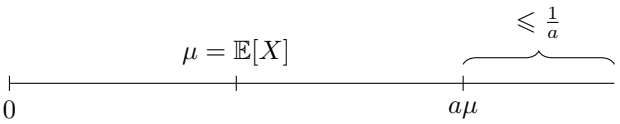


Figure 4: Markov's inequality: the probability that X is at least $a\mu$ is at most $1/a$.

Proof. Let

$$I = \begin{cases} 1 & \text{if } X \geqslant b, \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$I \leqslant \frac{X}{b}.$$

This is true for $X \geqslant b$, since then $I = 1 \leqslant \frac{X}{b}$. It is also true for $X < b$, since then $I = 0$ and $X \geqslant 0$. Markov's inequality follows by taking expectations on both sides:

$$\mathbb{E}[I] \leqslant \mathbb{E}\left[\frac{X}{b}\right] = \frac{\mathbb{E}[X]}{b}.$$

Since

$$\mathbb{P}(I = 1) = \mathbb{P}(X \geqslant b) \qquad \text{and} \qquad \mathbb{P}(I = 0) = \mathbb{P}(X < b),$$

we have $\mathbb{E}[I] = \mathbb{P}(I = 1)$, and hence

$$\mathbb{P}(X \geqslant b) \leqslant \frac{\mathbb{E}[X]}{b}.$$

Taking $b = a\,\mathbb{E}[X]$ gives $\mathbb{P}(X \geqslant a\,\mathbb{E}[X]) \leqslant \frac{1}{a}$. □

7 Chebyshev's Inequality

Chebyshev's inequality uses the variance of the random variable, which is also called the *second moment*.

Definition 2 ▶ Variance and Standard Deviation

The **variance** of a random variable X is

$$\mathbb{V}[X] = \mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big].$$

The **standard deviation** of X is

$$\sigma = \sqrt{\mathbb{V}[X]}.$$

Theorem 6 ▶ Chebyshev's Inequality

For any random variable X and any $a > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geqslant a) \leqslant \frac{\mathbb{V}[X]}{a^2} = \frac{\sigma^2}{a^2}.$$

Here $|X - \mathbb{E}[X]|$ is the deviation of X from its mean.

Proof.

$$\mathbb{P}(|X - \mathbb{E}[X]| \geqslant a) = \mathbb{P}\Big((X - \mathbb{E}[X])^2 \geqslant a^2\Big) \leqslant \frac{\mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big]}{a^2} = \frac{\mathbb{V}[X]}{a^2},$$

where the inequality is Markov's inequality applied to the random variable $(X - \mathbb{E}[X])^2 \geqslant 0$. □

8 Chernoff Bound

Let

$$X = \sum_{i=1}^{n} X_i, \qquad X_i = \begin{cases} 1 & \text{with probability } p_i, \\ 0 & \text{with probability } 1 - p_i, \end{cases}$$

where the X_i are *mutually independent*. By linearity of expectation,

$$\mathbb{E}[X] = \mu = \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} p_i.$$

Theorem 7 ▶ Chernoff Bound

Let X and μ be as above. Then

- for $0 < \delta < 1$,
$$\mathbb{P}(X \geqslant (1+\delta)\mu) \leqslant e^{-\frac{\mu\delta^2}{3}};$$
- for $R \geqslant 6\mu$,
$$\mathbb{P}(X \geqslant R) \leqslant 2^{-R};$$
- for $0 < \delta < 1$,
$$\mathbb{P}(X < (1-\delta)\mu) \leqslant e^{-\frac{\mu\delta^2}{2}}.$$

9 Search

Let $f\colon D \to R$ be a function from a domain D to a range R (Figure 5). Given y, we ask: is there an x such that $f(x) = y$? We consider

$$f\colon [1, \dots, N] \to [1, \dots, N],$$

and we want to find an x such that $f(x) = k$. The function can be seen as an array with n slots, where n is the size of the domain, and the cost of an algorithm is the number of queries to f.

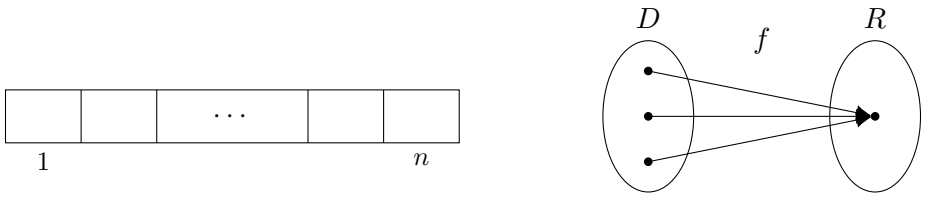


Figure 5: Left: an array with n slots. Right: a function $f\colon D \to R$.

- Randomized classical algorithm: $\Theta(n)$ queries.
- Grover's algorithm: $\mathcal{O}(\sqrt{n})$ queries.

For the random sampling algorithm,

$$\mathbb{E}[\text{queries}] = \mathcal{O}(n), \qquad \mathbb{P}(\text{Success}) = \frac{1}{n}.$$

Probability amplitude amplification. If the classical success probability is p, then in the quantum case it becomes $\sqrt{p}$.

10 Birthday Paradox

There are m people and n birthdays ($n = 365$). We are looking for a *collision*, i.e., $x \neq y$ such that

$$f(x) = f(y),$$

where $f(x)$ is the birthday of person x.

Let E_i be the event that the ith birthday is different from the first $i-1$ birthdays. Then

$$\begin{aligned}
\mathbb{P}\left(\bigcap_{i=1}^{m} E_i\right) &= \mathbb{P}(\text{all birthdays different}) \\
&= \mathbb{P}(E_1) \cdot \mathbb{P}(E_2 \mid E_1) \cdot \mathbb{P}(E_3 \mid E_2 \cap E_1) \cdots \\
&= 1 \cdot \left(1 - \frac{1}{n}\right) \cdot \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{m-1}{n}\right) \\
&= \prod_{i=0}^{m-1} \left(1 - \frac{i}{n}\right) \\
&\leqslant \prod_{i=0}^{m-1} e^{-i/n} \\
&= e^{-\sum_{i=0}^{m-1} \frac{i}{n}} \\
&= e^{-\frac{(m-1)m}{2n}}.
\end{aligned}$$

For $m = \sqrt{2n} + 1$,

$$\mathbb{P}(\text{all birthdays different}) \leqslant e^{-\frac{\sqrt{2n}(\sqrt{2n}+1)}{2n}} \leqslant e^{-\frac{2n}{2n}} = \frac{1}{e} \leqslant \frac{1}{2}.$$

Hence $\mathcal{O}(\sqrt{n})$ people are enough to have a constant probability of collision.

10.1 Expected Number of Collisions

The birthday paradox can be seen as throwing m balls into n bins (Figure 6). We show that for $m \approx \sqrt{2n}$,

$$\mathbb{E}[\#\text{ collisions}] \geqslant 1.$$

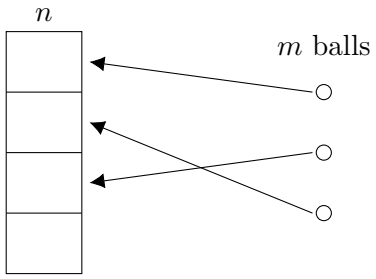


Figure 6: m balls are thrown into n bins.

Let X be the number of collisions. X takes values from 0 to $\binom{m}{2}$. Write

$$X = \sum_{i \neq j} X_{ij}, \qquad X_{ij} = \begin{cases} 1 & \text{if } f(i) = f(j), \\ 0 & \text{else,} \end{cases}$$

where the sum is over the $\binom{m}{2}$ pairs $i \neq j$. The random variables X_{ij} are *pairwise independent*. By linearity of expectation,

$$\begin{aligned} \mathbb{E}[X] &= \sum_{i \neq j} \mathbb{E}[X_{ij}] \\ &= \sum_{i \neq j} \mathbb{P}(X_{ij} = 1) \\ &= \sum_{i \neq j} \frac{1}{n} \\ &= \binom{m}{2} \frac{1}{n} = \frac{m(m-1)}{2n} \sim \frac{m^2}{n}. \end{aligned}$$

Exercise 2. Show that

$$\mathbb{P}(X \geqslant 1) \geqslant \Theta(1),$$

i.e., the birthday paradox occurs.

From class:

$$\mathbb{V}[X] = \mathbb{V}\left[\sum_{i \neq j} X_{ij}\right].$$

Solution.

11 Quantum Birthday Paradox

An algorithm makes k queries

$$f(x_1), \ldots, f(x_k).$$

Exercise 3. Show a $\sqrt{n}$ lower bound in the classical case.

Solution.